**Effat College of Engineering · Computer Science**

# CS3081 Artificial Intelligence
# Assignment: Uncertainty and Bayesian Networks

**Instructor:** Dr. Naila Marir **Total:** 50 marks (5 exercises × 10 marks)

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## Instructions

1. Show all your steps. If you write only the final number, you get at most half of the marks.
2. Write probabilities with 4 decimal places (for example 0.5367), unless the question says something else.
3. Exercises 1 to 4, and Exercise 5 parts (a) and (b), are solved by hand.
4. Exercise 5 part (c) is solved in Python with pgmpy in VS Code, like in the lab.
5. Submit two files on Blackboard: these pages with your answers written in the boxes, scanned as one PDF, and your notebook `assignment_bn.ipynb` with all outputs.
6. This is individual work. You can discuss ideas with classmates, but you must write your own answers.

## Exercise 1 — Joint distribution (10 marks)

The Effat shuttle bus is either **on time** or **late**. The weather in Jeddah is either **clear** or **dusty**. The table gives the probability of each pair. For example, the probability that the weather is clear **and** the bus is late is 0.14.

| Weather | Shuttle = on time | Shuttle = late |
|---|---|---|
| clear | 0.56 | 0.14 |
| dusty | 0.12 | 0.18 |

1. **(2 marks)** Show that the four numbers in the table make a valid probability distribution.

$0.56 + 0.14 + 0.12 + 0.18 = 1$

All probabilities are between 0 and 1 and their sum is 1.

Therefore, it is a valid probability distribution.

Conditions:

① $0 \leq \Sigma P \leq 1$

② $\Sigma P = 1$

2. **(2 marks)** Find P(dusty), the probability that the weather is dusty. Then find P(late), the probability that the bus is late.

Marginalization: $P(a) = P(a,b) + P(a,\sim b)$

→ $P(\text{dusty}) = 0.12 + 0.18 = 0.3000$

→ $P(\text{late}) = 0.14 + 0.18 = 0.3200$

3. **(2 marks)** Find P(late | dusty) and P(late | clear).

using Conditional probability : $P(a|b) = \frac{P(a,b)}{P(b)}$

→ $P(\text{late} \mid \text{dusty}) = \frac{0.18}{0.30} = 0.6000$

→ $P(\text{late} \mid \text{clear}) = \frac{0.14}{0.70} = 0.2000$

$P(\text{clear}) = 0.56 + 0.14 = 0.70$

4. **(2 marks)** Find P(dusty | late). Is it the same as P(late | dusty)? Explain your answer in one sentence.

$P(\text{dusty} \mid \text{late}) = \frac{P(\text{dusty, late})}{P(\text{late})} = \frac{0.18}{0.32} = 0.5625$

No, they're not the same. $P(\text{dusty} \mid \text{late}) = 0.5625$ & $P(\text{late} \mid \text{dusty}) = 0.6000$

because the conditioning events are different.

5. **(2 marks)** Are the weather and the bus independent? Use numbers to explain your answer.

For independence : $P(\text{dusty, late}) = P(\text{dusty})\,P(\text{late})$

$P(\text{dusty, late}) = 0.1800$

$P(\text{dusty})\,P(\text{late}) = 0.30(0.32) = 0.0960$

$0.1800 \neq 0.0960$ Therefore the weather & shuttle are not independent.

## Exercise 2 — Bayes' rule: a spam filter (10 marks)

An email program wants to find spam emails. It knows these facts:

- 30% of all emails are spam. So P(spam) = 0.3 and P(normal) = 0.7.
- In spam emails, the word "free" appears 60% of the time: P(free | spam) = 0.6.
- In normal emails, the word "free" appears 5% of the time: P(free | normal) = 0.05.

1. **(3 marks)** Find P(free), the probability that any email contains the word "free". Write the name of the rule you use.

Rule : Conditioning

$P(\text{free}) = P(\text{free} \mid \text{spam})\,P(\text{spam}) + P(\text{free} \mid \text{normal})\,P(\text{normal})$

$= (0.60)(0.30) + (0.05)(0.70)$

$= 0.1800 + 0.0350$

$P(\text{free}) = 0.2150$

2. **(3 marks)** A new email contains the word "free". Use Bayes' rule to find P(spam | free).

Bayes' rule

$$P(\text{spam} \mid \text{free}) = \frac{P(\text{free} \mid \text{spam})\,P(\text{spam})}{P(\text{free})}$$

$$= \frac{(0.60)(0.30)}{0.2150} = \frac{0.1800}{0.2150} = 0.8372$$

3. **(3 marks)** The program also checks the word "winner": P(winner | spam) = 0.4 and P(winner | normal) = 0.02. Assume that, once we know if an email is spam or normal, the two words appear independently of each other. A new email contains **both** "free" and "winner". Find P(spam | free, winner).

$$P(\text{free}, \text{winner} \mid \text{spam}) = 0.6(0.4) = 0.24$$

$$P(\text{free}, \text{winner} \mid \text{normal}) = 0.05(0.02) = 0.001$$

$$P(\text{spam} \mid \text{free}, \text{winner}) = \frac{0.24(0.3)}{0.24(0.3) + 0.001(0.7)} = \frac{0.072}{0.0727} = 0.9904$$

4. **(1 mark)** Compare your answers to parts 2 and 3. Did the probability of spam go up or down when we added the word "winner"? Explain why in one sentence.

The probability increased from 0.8372 to 0.9902 because "winner" is more likely in spam emails.

## Exercise 3 — Bayesian network: structure and joint probability (10 marks)

In Jeddah summers, a **heatwave** (H) can cause an **AC overload** (A) in a campus building. A heatwave and an AC overload can both cause a **power cut** (C). A power cut can cause the **lab session to be cancelled** (L). Each variable is yes or no.

The network has 4 arrows: H → A, H → C, A → C, and C → L. The tables give only the probability of "yes". The probability of "no" is 1 minus the probability of "yes".

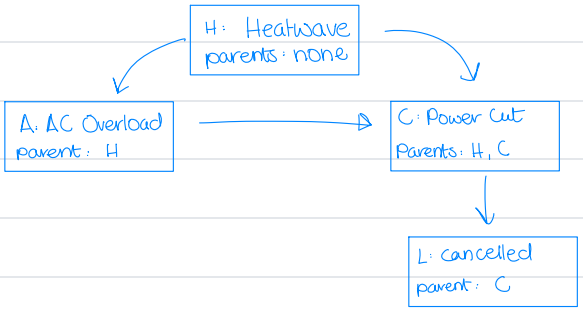
| P(H = yes) |
|---|
| 0.3 |

| H | P(A = yes \| H) |
|---|---|
| yes | 0.6 |
| no | 0.1 |

| H | A | P(C = yes \| H, A) |
|---|---|---|
| yes | yes | 0.7 |
| yes | no | 0.2 |
| no | yes | 0.5 |
| no | no | 0.05 |

| C | P(L = yes \| C) |
|---|---|
| yes | 0.8 |
| no | 0.1 |

1. **(2 marks)** Draw the network. For each node, write its parents.




2. **(2 marks)** Write P(H, A, C, L) as a product of the probabilities from the network's tables.

$P(H,A,C,L) = P(H)P(A|H)P(C|H,A)P(L|C)$

3. **(2 marks)** A full joint table for 4 yes/no variables needs 15 numbers ($2^4 - 1$). How many numbers do the 4 tables of this network need? Count only the "yes" probabilities, and show how you counted.

$H=1, \quad A=2, \quad C=4, \quad L=2$

$1 + 2 + 4 + 2 = 9$

4. **(2 marks)** Find P(H = yes, A = no, C = yes, L = yes).

$P(H=y, A=n, C=y, L=y)$

$= 0.3(0.4)(0.2)(0.8) = 0.0192$

5. **(2 marks)** Find P(H = yes | C = yes). Then give P(H = yes | C = yes, L = yes) without doing a new calculation, and explain why the two answers are the same.

$P(H=y, C=y) = 0.3[0.6(0.7) + 0.4(0.2)] = 0.1500$

$P(H=n, C=y) = 0.7[0.1(0.5) + 0.9(0.05)] = 0.0665$

$P(H=y \mid C=y) = \frac{0.1500}{0.1500 + 0.0665} = 0.6928$

$P(H=y \mid C=y, L=y) = 0.6928$ Both are equal because L depends only on C.

## Exercise 4 — Inference by enumeration, challenge question (10 marks)

This is the **challenge question** of the assignment. Show every step.

Use the network and tables from Exercise 3. Today the lab session **was cancelled** (L = yes).

1. **(4 marks)** Use inference by enumeration to find P(H = yes | L = yes), the probability that there was a heatwave. Write the query variable, the evidence and the hidden variables, then show all sums and the normalization.

P(H = yes | L = yes)

Query: H

Evidence: L = yes

Hidden variables: A, C

$$P(H \mid L=y) = \alpha \sum_A \sum_C P(H, A, C, L=y)$$

For H = yes: $P(H=y, L=y) = 0.3(0.6)(0.7)(0.8) + 0.3(0.6)(0.3)(0.1) + 0.3(0.4)(0.2)(0.8) + 0.3(0.4)(0.8)(0.1) = 0.1008 + 0.0054 + 0.0192 + 0.0096 = 0.1350$

For H = no:

$P(H=n, L=y) = 0.7(0.1)(0.5)(0.8) + 0.7(0.1)(0.5)(0.1) + 0.7(0.9)(0.05)(0.8) + 0.7(0.9)(0.95)(0.1) = 0.0280 + 0.0035 + 0.0252 + 0.05985 = 0.11655$

Normalization:

$\alpha = \frac{1}{0.1350 + 0.11655} = \frac{1}{0.25155}$

$P(H=y \mid L=y) = \frac{0.1350}{0.25155} = 0.5367$

2. **(3 marks)** Now find P(A = yes | L = yes), the probability that there was an AC overload. Which variables are hidden this time?

Query: A

Evidence: L = yes

Hidden variables: H, C

$$P(A \mid L=y) = \alpha \sum_H \sum_C P(H, A, C, L=y)$$

$P(A=y, L=y) = 0.1008 + 0.0054 + 0.0280 + 0.0035 = 0.1377$

$P(A=n, L=y) = 0.0192 + 0.0096 + 0.0252 + 0.05985 = 0.11385$

$P(A=y \mid L=y) = \frac{0.1377}{0.1377 + 0.11385} = 0.5474$

3. **(3 marks)** A technician checks the building and reports that there was **no** AC overload (A = no). The lab is still cancelled (L = yes). Find P(H = yes | L = yes, A = no). Compare it with your answer to part 1: did the probability of a heatwave go up or down? Use the network to explain why in two or three sentences.

Query: H

Evidence: L = yes, A = no

Hidden variable: C

$P(H \mid L=y, A=n) = \alpha \sum_C P(H, A=n, C, L=y)$

$P(H=y, A=n, L=y) = 0.0192 + 0.0096 = 0.0288$

$P(H=n, A=n, L=y) = 0.0252 + 0.05985 = 0.08505$

$P(H=y \mid L=y, A=n) = \frac{0.0288}{0.0288 + 0.08505} = 0.2530$

The probability decreased from 0.5367 to 0.2530. NO AC overload makes a heatwave less likely because heatwaves increased the probability of an overload.

## Exercise 5 — Approximate inference by sampling (10 marks)

Now we find the same probability, P(H = yes | L = yes), by **sampling**. Sampling gives an estimate, not the exact answer.

**(a) Rejection sampling (3 marks).** Reminder: in rejection sampling, we keep only the samples that agree with the evidence (here L = yes), and we throw away the others. Below are 12 samples from the network of Exercise 3.

| # | H | A | C | L |
|---|---|---|---|---|
| 1 | no | no | no | no |
| 2 | yes | yes | yes | yes |
| 3 | no | no | no | no |
| 4 | no | yes | yes | yes |
| 5 | yes | yes | no | no |
| 6 | no | no | no | yes |
| 7 | yes | no | yes | yes |
| 8 | no | no | no | no |
| 9 | no | no | no | no |
| 10 | yes | yes | yes | no |
| 11 | no | no | no | no |
| 12 | no | no | yes | yes |

1. Write the numbers of the samples we keep and the samples we throw away. Then estimate P(H = yes | L = yes).

Keep : 2, 4, 6, 7, 2

Reject : 1, 3, 5, 8, 9, 10, 11

$P(H = y \mid L = y) \approx \frac{2}{5} = 0.4000$

2. What fraction of the 12 samples did we throw away? Why is this a problem for rejection sampling?

$\frac{7}{12} = 0.5833$

Rejection sampling wastes samples that do not match the evidence, reducing efficiency.

**(b) Likelihood weighting (3 marks).** Reminder: in likelihood weighting, we do not sample the evidence. We fix it (here L = yes), we sample the other variables, and we give each sample a **weight**. The weight is the probability of the evidence given the sample's values: here, P(L = yes | C). Below are 6 samples.

| # | H | A | C | L |
|---|---|---|---|---|
| 1 | no | no | no | yes (fixed) |
| 2 | yes | yes | yes | yes (fixed) |
| 3 | no | no | no | yes (fixed) |
| 4 | yes | no | no | yes (fixed) |
| 5 | no | yes | yes | yes (fixed) |
| 6 | yes | yes | yes | yes (fixed) |

1. Write the weight of each sample.

Sample → weight

1 → 0.1
2 → 0.8
3 → 0.1
4 → 0.1
5 → 0.8
6 → 0.8

2. Estimate P(H = yes | L = yes) using the weights: add the weights of the samples where H = yes, then divide by the sum of all weights.

$P(H=y \mid L=y) \approx \frac{0.8+0.1+0.8}{0.1+0.8+0.1+0.1+0.8+0.8} = \frac{1.7}{2.7} = 0.6296$

3. Compare your estimates from (a) and (b) with the exact answer from Exercise 4. Why are they different? How can we make them closer to the exact answer?

Exact = 0.5367, rejection sampling = 0.4000, likelihood weighting = 0.6296
The estimates differ because of the sample sizes
Increasing the no of samples generally improves accuracy.

**(c) Python with pgmpy (4 marks).** Answer this part in your notebook, not on paper. In VS Code, create `assignment_bn.ipynb`.

1. Build the network of Exercise 3 with `DiscreteBayesianNetwork` and `TabularCPD`, like in the lab. Name the nodes `heatwave`, `overload`, `powercut` and `cancel`, and use the values `yes` and `no`. Print `model.check_model()`. It must print `True`.
2. Use `VariableElimination` to find P(H | L = yes). Your answer must be the same as in Exercise 4.
3. Use rejection sampling with 10,000 samples to estimate P(H = yes | L = yes). Hint: import `BayesianModelSampling` from `pgmpy.sampling` and `State` from `pgmpy.factors.discrete`. Then use `BayesianModelSampling(model).rejection_sample(evidence=[State("cancel", "yes")], size=10000)`. This gives a table of samples. Count the fraction of rows where `heatwave` is `yes`.
4. In a Markdown cell, write two sentences that compare the exact answer and the sampling estimate.